

module and enter the file name as EFYTD. Do not type any port names here, just click 'Next.' The source EFYTD.vhd file is created. Now click on the 'Finish' button to complete the wizard (Fig. 4).

5. On the code page you have to type the code of the article. Entity should be EFYTD in the code. After entering it, select 'Behavioural simulation' in the 'Sources' window. In the 'Processes' window, select the 'Xilinx ISE simulator' option, double click on 'Check syntax,' and then on 'Run' option (Fig. 5).

6. Now create a test bench using 'Create New Source.' Select the VHDL test bench and enter the file name, say

'efytd.' Then click on the 'Next' button; the test bench file is created automatically. Select 'Test bench file' and 'Behavioural simulation' in the 'Sources' window. In the 'Processes' window, select 'Xilinx ISE simulator' and the 'Check syntax' option.

7. Once syntax checking is 'ok' (indicated by a green coloured circle), select the 'Simulate' option and run the program. The program output in Fig. 6 shows the multiplexed wave on the 'dout' output signal and Fig. 7 shows the internal signals s1, s2, s3 and s4.

8. You can add the test bench waveform in a similar manner by selecting 'Create New Source,' 'Execute' and the

'VHDL test bench waveform' options followed by entering the file name, say, 'efytd.v.' Set the waveform properties at default values unless it is otherwise necessary. Then select the Xilinx ISE simulator. Run the behavioural simulation for the waveforms.

The above steps for testing the simulated time division multiplexer have been successfully implemented. Improved ISE Xilinx 9.2i permits great flexibility for design and simulation.

EFY note. The source code and relevant files of this project have been included in this month's EFY-CD.

Both the authors are lecturers at the Government Polytechnic Bidar, Karnataka

EFYTD.VHD

```
--TDM Simulator
-- signal generator
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_ARITH.ALL;
use IEEE.STD_LOGIC_UNSIGNED.ALL;

entity sig_gen is
generic (hperiod : time);
port(s:inout std_logic;
en : in bit);
end entity sig_gen;
architecture arch_comp of sig_gen is
begin
process
begin
if en = '1' then
s <= '1';
wait for hperiod;
s <= '0';
wait for hperiod;
end if;
end process;
end architecture arch_comp;

-- Main unit TDM
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

use IEEE.STD_LOGIC_ARITH.ALL;
use IEEE.STD_LOGIC_UNSIGNED.ALL;

entity EFYTD is
port(din : inout std_logic_vector(3
downto 0);
sel : inout std_logic_vector(1 downto
0);
dout: out std_logic);
end EFYTD;

architecture Behavioral of EFYTD is
component sig_gen is
generic (hperiod : time);
port(s:inout std_logic;
en : in bit);
end component sig_gen;
signal s1,s2,s3,s4,mclk : STD_LOGIC;
begin
sg1: sig_gen generic map (2ns)
port map (s1,'1');
sg2: sig_gen generic map (4ns)
port map (s2,'1');
sg3: sig_gen generic map (8ns)
port map (s3,'1');
sg4: sig_gen generic map (16ns)
port map (s4,'1');

mstrclk: sig_gen generic map (30ns)
begin
port map (mclk,'1');
sels: process(mclk)
variable temp:integer:=0;
begin
if mclk = '1' and mclk'event then
temp := temp +1;
end if;
sel<= CONV_STD_LOGIC_VECTOR(temp,2);
end process sels;

mainp:process(s1,s2,s3,s4,sel,din,mclk)
begin
din(0)<= s1 ;
din(1)<= s2;
din(2)<= s3;
din(3)<= s4;
case sel is
when "00" => dout<= din(0) ;
when "01" => dout<= din(1);
when "10" => dout<= din(2);
when "11" => dout<= din(3);
WHEN OTHERS => dout <= 'U';
end case;
end process mainp;
end Behavioral;
```

QUIZ PROGRAM IN JAVASCRIPT

■ BUKYA SRIDHAR

Computerisation has touched almost every sector today. Computer-based software

programs are widely used for gaming, quizzes, etc. This application of computers can be extended to conduct computer-based tests (CBT) for various competitive exams or for on-line test-

ing. It eventually quickens the process of reviewing the tests and generating results.

In India, most competitive exams have moved from entering the cor-

